

INTERNSHIP SUMMARY REPORT

on the activities carried out during the internship

Host organisation S.A. "Orange Systems"

Faculty of *Computers, Informatics, and Microelectronics*

Study programme *Software Engineering*

Mode of study *full time*

Year of study: 3 **Group:** FAF-243

Internship period: from 01.09.2026 to 26.09.2026

Activities carried out and results obtained:

Activities carried out:

During my internship at Orange Systems, I worked on the SecretShare application, focusing heavily on infrastructure, CI/CD pipelines, and authentication integration. I configured automated ephemeral preview deployments using GitHub Actions and Tailscale SSH. I containerized the API using a multi-stage Dockerfile and orchestrated the deployment stack using Docker Compose with Traefik acting as the reverse proxy. I also integrated Keycloak for authentication, setting it up to persist data in PostgreSQL and auto-provision realms with test users. On the frontend, I migrated the application to Auth.js, ensuring proper subpath routing and implementing OAuth2 with PKCE support. I updated the project architecture documentation. While managing these configurations, I also referenced a file named temp.txt. Finally, I established an automated CI/CD workflow to build and deploy the project's LaTeX report.

Results obtained:

My contributions resulted in a robust CI/CD pipeline that automatically spins up preview environments for pull requests. The application now features a zero-configuration Tailscale staging environment utilizing stable path-based routing. Keycloak authentication is fully functional and successfully persists its state. Additionally, the frontend and backend services are properly dockerized with integrated health checks, and the team's LaTeX documentation is deployed automatically without manual intervention.

Difficulties encountered during the internship:

One major difficulty was configuring Keycloak and Traefik routing, specifically resolving issuer URL mismatches by enforcing `X-Forwarded-Port=443` and `X-Forwarded-Proto` headers to properly strip port 80. I also encountered Next.js Server-Side Rendering (SSR) white-page issues when attempting to trigger Keycloak logins via Server Actions, which required reverting the approach and adjusting the Auth.js route handlers. Another critical challenge was Keycloak repeatedly crashing due to Out-Of-Memory (OOM) kills; I resolved this by explicitly increasing its JVM memory limit to 1GB. Finally, handling git worktree synchronizations on the deployment server required writing robust scripts to prevent reference collisions during PR updates.

Conclusions and recommendations:

This internship bridged the gap between application development and DevOps platform engineering. I learned that configuring a reverse proxy and securing authentication flows requires deep attention to HTTP headers, environment variables, and proxy routing rules. I recommend that future teams test their reverse proxy and subpath configurations as early as possible to catch elusive URL rewriting issues. The company provided a highly supportive environment, and the mentorship was critical in helping me navigate these complex infrastructure and CI/CD challenges. Overall, the experience significantly strengthened my technical foundation in deployment automation and infrastructure stability.

Date of completion: 26/09/2026

Signature of the student: Covali Ilie